

INTRODUCTION

Current EMS cot designs provide inherent risks to patients and providers. This risk is caused by the cot’s **static design** and **small base** which results in a high likelihood of the cot overturning on uneven terrain. The instability creates a situation where EMS responders must **provide compensation to offset the cot’s failures**, placing themselves at risk. **1 in 4 EMTs sustain career ending back injuries** within the **first 4 years of working** (EMS Safety, 2011). These injuries largely result from **moving patients** and **compensating** for traditional EMS cot **failures** on uneven terrain (Fratta et al., 2020).



Fig.1: Stryker EMS Cot
(Power-pro XT | Stryker, 2010)



Fig.2: 1958 Ferno "All Level Cot"
(Bourgraf E., 2010)

As outdoor sports continue to gain popularity, the need for **trail rescue** becomes greater. Due to large amounts of **instability**, the use of EMS cots is **limited to rescues on flat terrain**. For trail rescues, providers must utilize transportation methods such as **backboards** and **stokes baskets**. These methods of extrication put **all physical loads on providers**, as they are carrying the patient’s full weight. This method of patient transportation, just like EMS cot compensation, places **significant risk** on the providers, often leading to **back injuries**.

PURPOSE

This project aims to reduce the need for **stability compensation** of the cot by providers and expand the function of safe transportation methods to include **diverse terrain**. Reducing the physical **load** will allow providers to **solely focus** on their patient and ultimately provide **better care**. Additionally, a more stable transportation system will create an environment that is **safer for patients** too. By creating a system in which the transportation device can **self-level the patient section**, a **patient’s weight** is **evenly distributed** throughout the cots frame, reducing its likelihood of overturning.

Development of a Self-Leveling Patient Transportation System Using Inclinometer-Based Feedback Control

METHODS

Calculations and Coding

The sensors and servos tested individually. The physics equations derived.

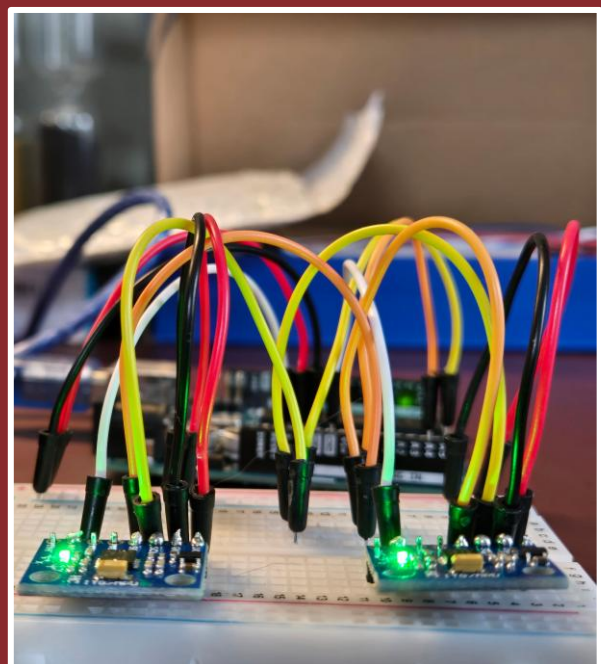


Fig.3: Inclinometer Testing
(Photo Taken by Student, 2026)



Fig.4: Simultaneous Servo Testing
(Photo Taken by Student, 2026)

Master Code created to incorporate all components and the calculations.

```
float pitch = atan(-AcX2 / sqrt(sq(AcY2)+sq(AcZ2-1.1 )));
float roll = atan(AcY2 / sqrt(sq(AcX2)+sq(AcZ2-1.1)));

float heightFR = 1.5*tan(pitch)+4.5*tan(roll);
val1 = atan(heightFR / sqrt(sq(1.5)+sq(4.5))) * RAD_TO_DEG;
```

Fig.5: In Code Physics Equations (Photo Taken by Student, 2026)

Fabrication and Assembly

Connective components designed in AutoCAD Inventor and 3D printed.

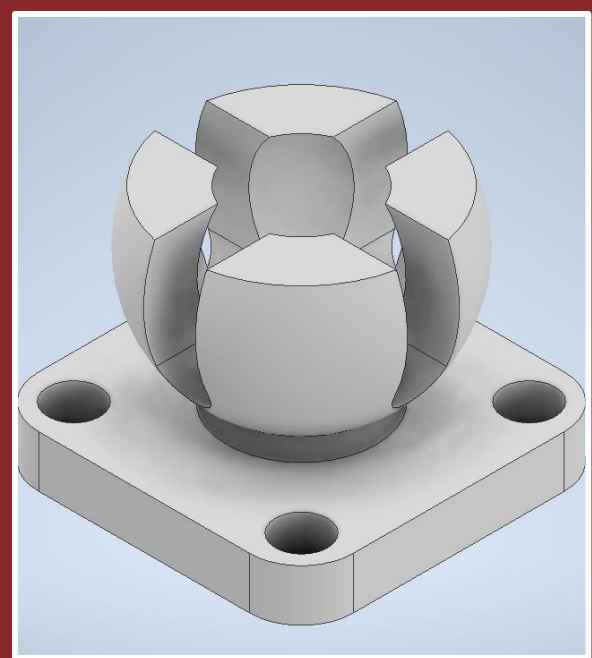


Fig.6: Ball Joint Socket CAD
(Photo Taken by Student, 2025)



Fig.7: Ball Joint Socket 3D Print
(Photo Taken by Student, 2025)

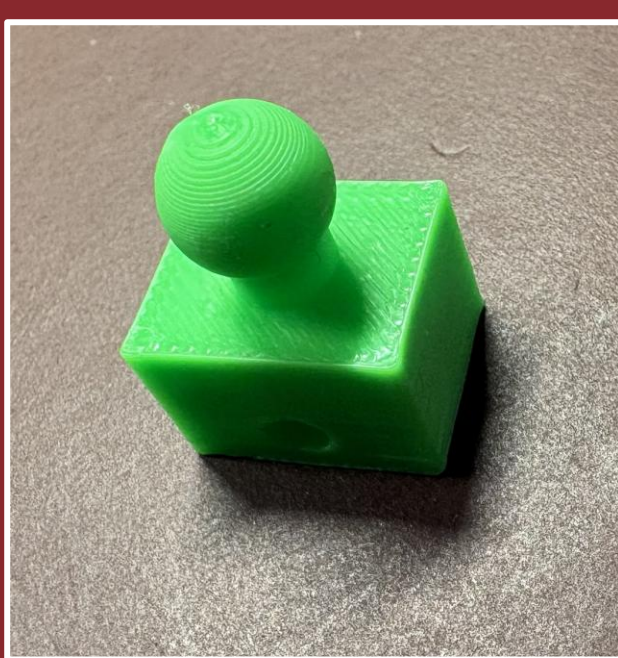


Fig.8: Ball Joint Socket 3D Print
(Photo Taken by Student, 2025)

Testing apparatus assembled and all component mounted.

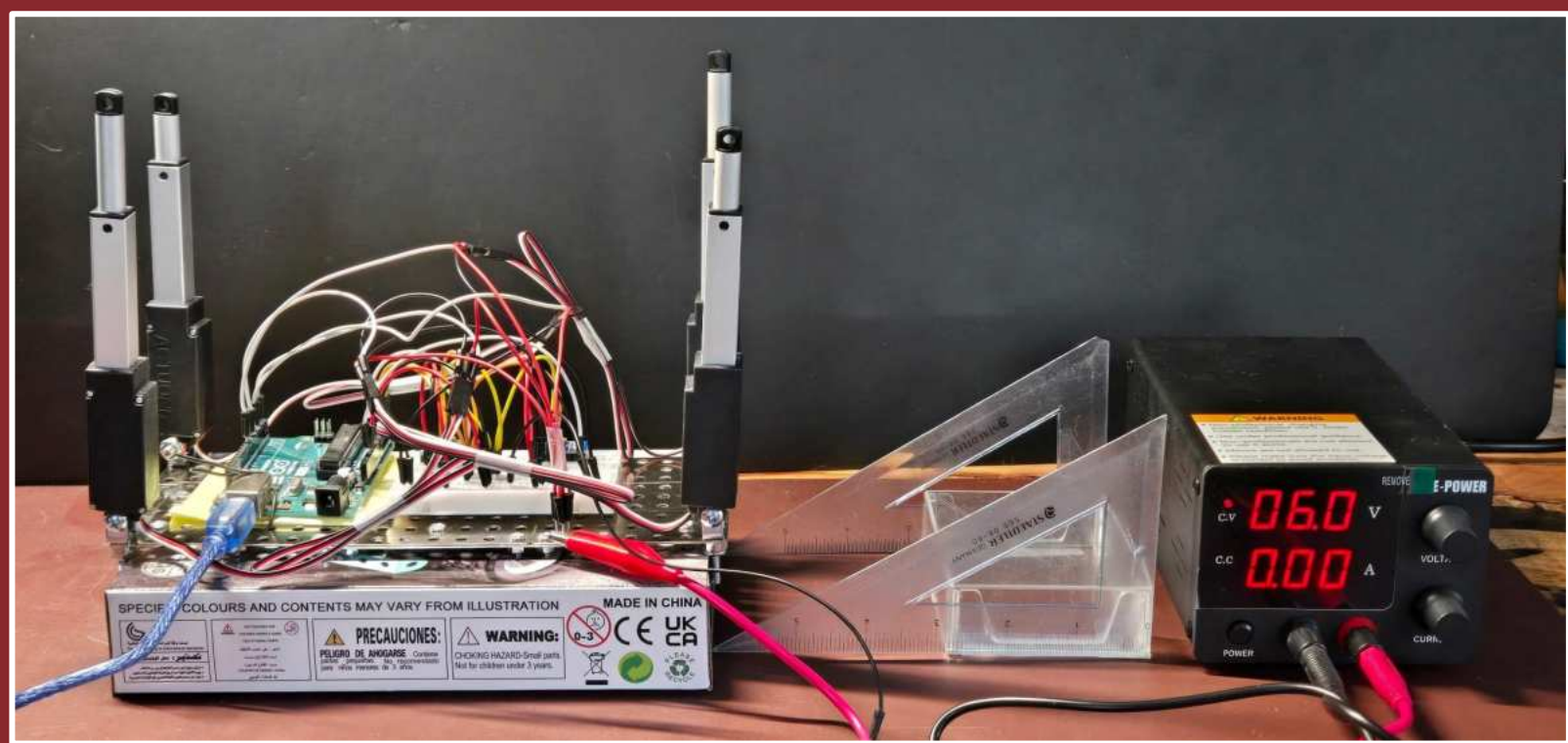


Fig.9: Testing Apparatus (Photo Taken by Student, 2026)

TESTING/RESULTS

Angle/Position	Y – Axis View Angle of Error	X – Axis View Angle of Error
-15 Degrees in X - Axis	2.2	4.2
-15 Degrees in Y - Axis	4.1	0.0
15 Degrees in (X,Y)	8.9	8.8
7.5 Degrees in X - Axis	3.0	0.0
-7.5 Degrees in Y - Axis	1.5	3.5

Testing included obtaining video recordings from the front and side perspectives as the prototype is placed through a series of specified angles. Video analysis was then utilized to quantify angle errors along both X and Y axes. The results indicate that the system was generally effective in maintaining near-level conditions under moderate angular displacement. The system, however, struggled to counter larger combined-axis displacement.

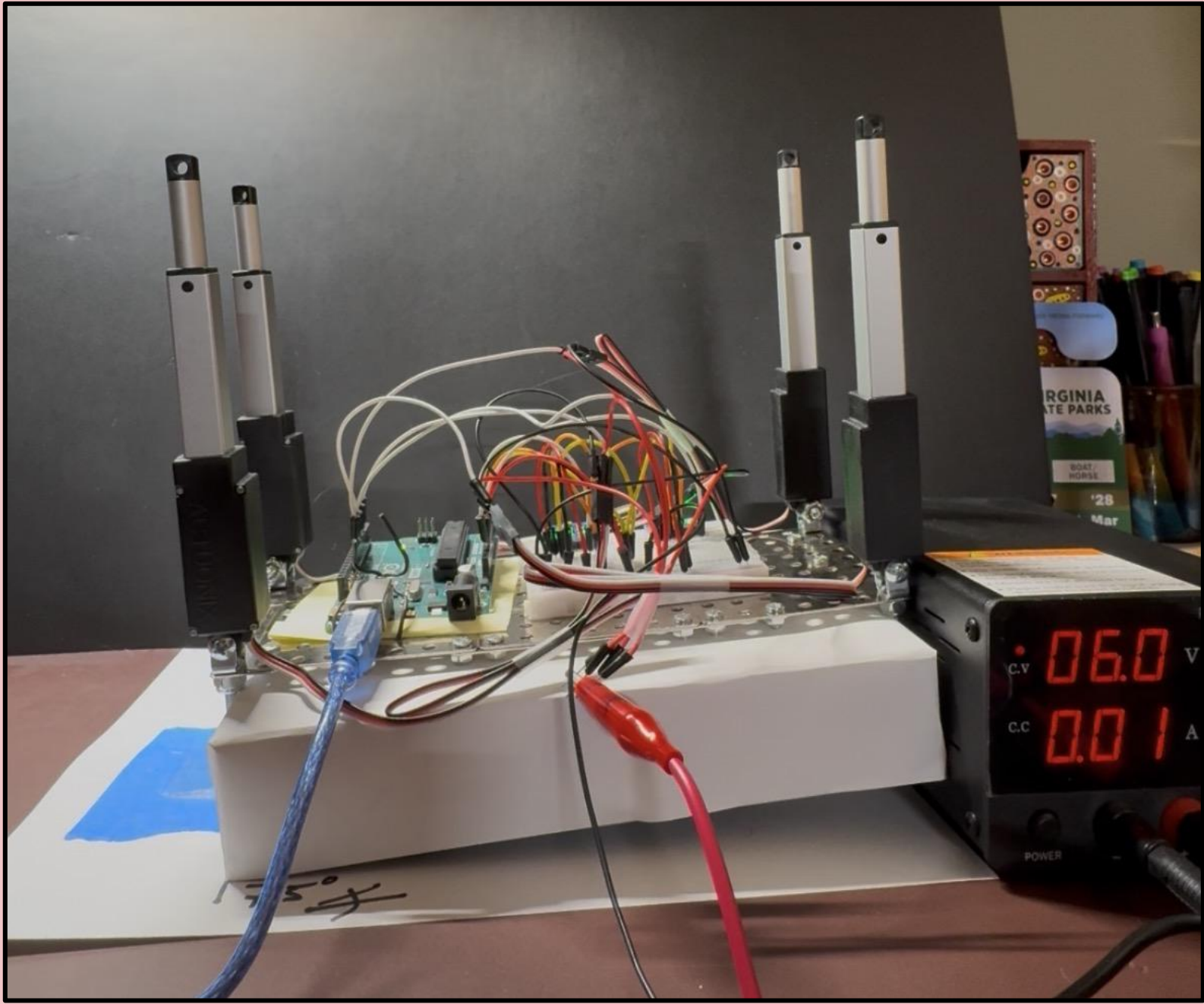


Fig.10: Testing (Photo Taken by Student, 2026)

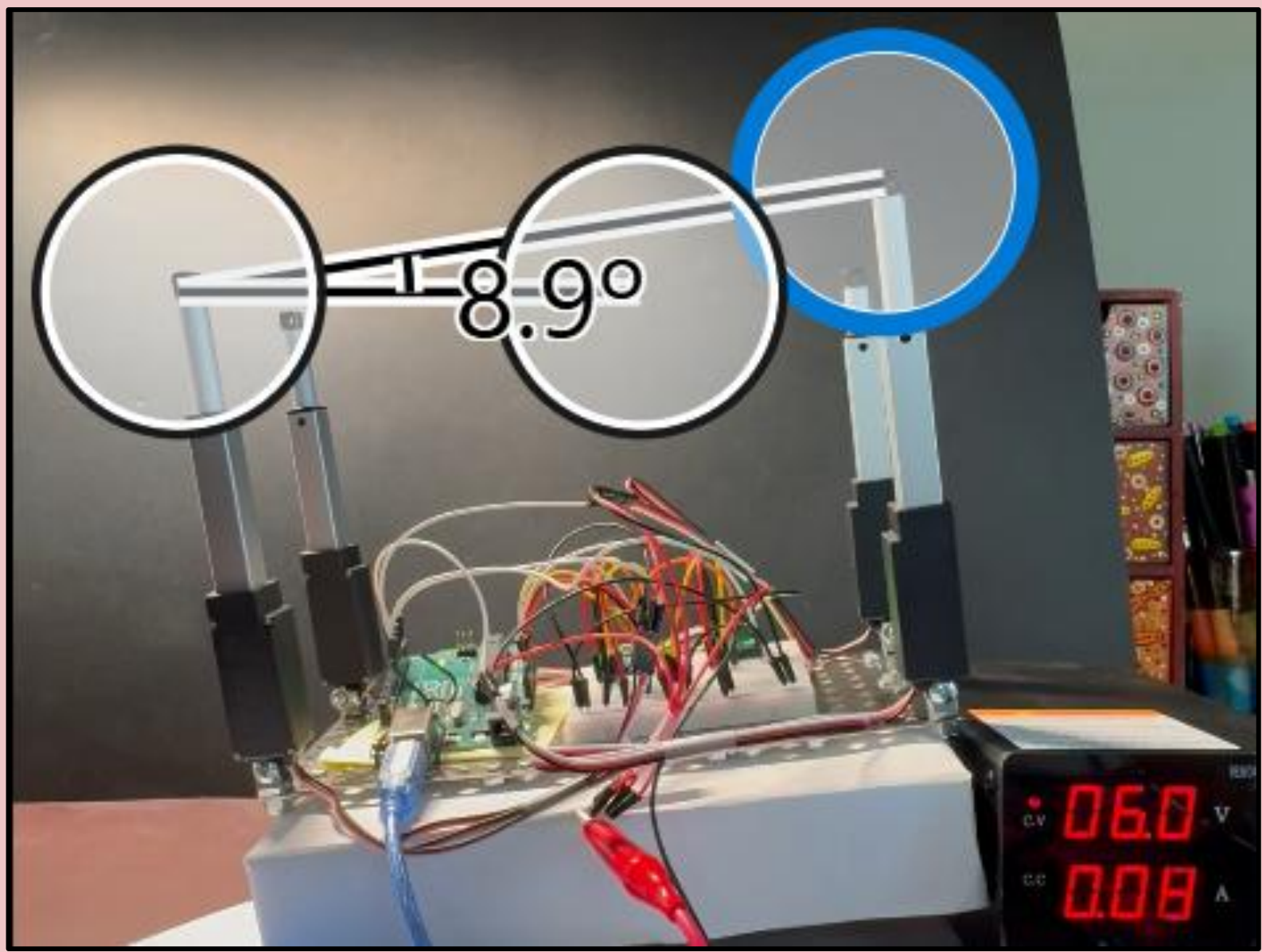


Fig.11: Video Analysis (Photo Taken by Student, 2026)

CONCLUSIONS

Final testing resulted in an **average angle error** of **3.94** along the Y-axis and **3.30** along the X-axis. The largest angle errors, for both the Y and X axes, occurred on the third test. In this test the **angles** on the Y and X axes were **manipulated concurrently**. This may indicate a remaining flaw in the final calculations or a need for a more stable testing surface. Further experimentation is required to truly determine where the issue stems from. Overall, an average angle error of 3.94 and 3.30 should not poorly impact a patient. While the technology has not yet been perfected, the **angle adjustment** that is occurring will still **provide some** much-needed **stability** to the transportation system. Ultimately, an angle of 3.94 degrees is significantly more stable and comfortable for a patient than 15 degrees.

EXTENSIONS

Further Research: Testing will be taken further to determine **load capacity** of the four-actuator system. Additionally, if the prototype seems viable, adjustments will be made to create a **larger scale prototype** the size of the typical EMS cot. Testing at the larger scale will provide the ability to **improve any potential design flaws** that may not have been noticeable on the smaller scale. Furthermore, creating a larger scale prototype allows **providers to verify results** and give feedback based on their observations and previous experience. **Other Applications:** This system could be implemented in other scenarios such as transportation of scientific equipment or construction materials. In the case of scientific equipment, keeping the materials as level as possible while moving will reduce errors later, by preventing internal components from shifting.

REFERENCES

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